

## **NIELIT Virtual Academy**

### **INTERSHIP PROSPECTUS**

**Name of the Internship:** Online Internship in RTL Design using Verilog HDL

**Internship Code:** IN13

**Mode of Conduction:** Online

**Starting Date:** 2<sup>nd</sup> July 2025

**Last date of registration:** 30<sup>th</sup> June 2025

**Duration:** 6 Weeks

### **Objective of the Course**

- ✓ The internship program aims to enable participants to design reusable Intellectual Property (IP) Cores as building blocks using Verilog HDL.

### **Outcome of the Course**

- ✓ Understand the brief history, present and future, and Design Cycle of VLSI technology.
- ✓ Learn and implement the HDL IPs using Verilog HDL from scratch.
- ✓ Emulate, debug & Characterize reusable IPs
- ✓ Provide hands-on for highly deterministic real-time applications

**Course Fee:** Rs: 1000/- (inclusive of GST)

**Eligibility:** Pursuing an Undergraduate level course or above

### **Methodology**

- ✓ Teaching Mode: Self-Pace
- ✓ Access from anywhere anytime
- ✓ Content Access through e-learning portal
- ✓ Doubt Removal Session
- ✓ Covers both Theory & Practical
- ✓ Certification: On completion of the Mini Project

**Registration Link:** <http://nva.nielit.gov.in>

## Contact Details:

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## Course Structure:

| Module No | Module Title                    |
|-----------|---------------------------------|
| 1         | VLSI Fundamentals               |
| 2         | RTL Design Methodology          |
| 3         | RTL Design Using HDL            |
| 4         | RTL Simulation and Verification |
| 5         | Timing Constraints and Analysis |
| 6         | Project                         |

## Syllabus:

### Detailed Syllabus

#### Module 1: VLSI fundamentals

- Overview of VLSI technology and its significance
- Introduction to VLSI design flow: Front-end and Back-end
- Fundamentals of MOSFET operation
- CMOS Inverter Characteristics
- CMOS Logic Design
- Transistor Level Schematics and Layouts
- On-Chip Wire Modeling
- Bonding Diagram, Packaging, and Assembly
- Combinational Logic Circuit Critical Path Optimization

#### Module 2: RTL Design Methodology

- Basics of Register Transfer Level (RTL) design
- Overview of RTL design process and methodology
- Introduction to Hardware Description Languages (HDLs) – Verilog

### **Module 3: RTL Design Using HDL**

- Introduction to Verilog syntax and construct
- Designing combinational and sequential logic using HDL
- Writing RTL code for basic digital circuits

### **Module 4: RTL Simulation and Verification**

- Functional verification techniques for RTL designs
- Introduction to test benches and test bench development
- Simulation and debugging of RTL designs

### **Module 5: Timing Constraints and Analysis**

- Introduction to timing constraints in RTL design
- Timing analysis and optimization techniques
- Setup and hold time violations and resolution

### **Module 6: Project**

- Capstone project
- Project Report Submission